



DEEP LEARNING TAKEAWAYS



Chapter:

Getting Started

Neural Networks: The Foundation of Deep Learning

- 1** Neural networks are inspired by the workings of the human brain and form the foundation of deep learning.
- 2** When features are complex (e.g., detecting whether an image is of a cat or a dog), we use deep learning. For simpler features and structured data, statistical machine learning is preferred.
- 3** Neural networks consist of input, hidden, and output layers. Neurons are responsible for detecting patterns in the data, working together like small processors on a larger task.
- 4** Deep learning is a machine learning technique that uses neural networks to learn from large amounts of data, mimicking the human brain's ability to recognize patterns and make decisions.

Deep Learning vs Statistical ML: When to Use What?

Criteria	Deep Learning	Statistical Machine Learning
Feature Complexity	Best for complex features (e.g., images, audio, text) where feature extraction is difficult or non-intuitive.	Best for simple or structured features (e.g., tabular data) where feature extraction can be done manually.
Interpretability	Low interpretability; neural networks often act as a "black box," making it harder to explain decisions.	High interpretability; models like decision trees or linear regression provide clearer insights into how decisions are made.
Data Volume	Requires large datasets for effective training due to the high number of parameters.	Can work effectively with smaller datasets, often performing well with fewer data points.
Compute Resources	Requires significant computational power (e.g., GPUs, TPUs) to train deep neural networks efficiently.	Less computationally intensive and can be trained using standard CPUs with lower resource requirements.

Neural Network Architectures

- 1 Just like how homes have different architectures (e.g. wood, metal, concrete), neural networks also have different architectures
- 2 Feed forward neural network, Recurrent neural network, Convolutional neural network, Transformers are some of the popular architectures

Real World Applications of DL

- 1** Deep learning is impacting our lives directly today. It has many real life use cases.
- 2** Here are some of the use cases of various neural network architectures,
 - a. Feed Forward Neural Network (FNN):
Weather prediction, demand forecasting
 - b. Convolutional Neural Network (CNN):
Autonomous driving, photo classification, disease diagnosis
 - c. Recurrent Neural Network (RNN):
Machine translation, Speech recognition (Google assistant)
 - d. Transformers: Gen AI (Using tools like ChatGPT)

Tooling: PyTorch vs Tensorflow

- 1 There are many opensource frameworks available today that allows you to build a deep learning system effectively
- 2 PyTorch (by Meta) and Tensorflow (by Google) are two popular frameworks. While both have their own benefits, PyTorch seems to be gaining popularity and it is preferred for students as the learning is easy compared to Tensorflow

Tooling: GPU, TPU

- 1** GPUs (Graphics Processing Units) were traditionally used for gaming but now they are being used for deep learning
- 2** NVIDIA is a company known for GPUs and the rise in their stock price is mainly attributed to Gen AI boom that has resulted into companies ordering millions of dollars of GPUs from NVIDIA
- 3** TPU stands for tensor processing unit. It is a less popular option used to speed up deep learning training
- 4** For this course, it is ok even if you do not have a GPU on your computer. If you have it then you will notice that training a model will be faster compared to CPU